



Anna Krystalli

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WORK EXPERIENCE

CEO & LEAD RESEARCH SOFTWARE ENGINEER – RSE SINGLE MEMBER PRIVATE COMPANY – 20/05/2022 – Current – ERMOUPOLI, SYROS, GREECE

Founded R-RSE in May 2022 to provide Research Software Engineering consultancy to academic and public-health research groups, specialising in R, reproducible research infrastructure, and community-maintained R packages.

SELECTED ENGAGEMENTS

- **Hubverse Modelling Hub Infrastructure:** multi-year engagement with the University of Massachusetts, Amherst (Prof. Nicholas Reich's group), in support of CDC-funded FluSight and related public-health forecasting. Core contributor across the hubverse R package ecosystem (validation, evaluation, data access, dashboard infrastructure) and founding work on AI tooling and knowledge infrastructure for the hubverse. Packages maintained on CRAN. <https://github.com/hubverse-org>
- **EMODnet Biology:** ongoing contributions under successive project phases. Delivered the emodnet.wcs R package and a series of four reproducible R tutorials launched publicly by EMODnet Biology. <https://emodnet.github.io/emodnet-bio-r-geo-tutorials/>
- **University of Bern:** designed and built an internal data validation Shiny application for upstream data quality control.

SELECTED TRAINING MATERIALS DEVELOPED & DELIVERED

- **Reproducible Research Data and Project Management in R:** five-day course delivered annually to NERC ACCE Doctoral Training Partnership 1st-year PhD students. Originated as a one-day workshop in 2015 and grown each year since; 2026 marked the 11th iteration. <https://acce-rrresearch-2026.netlify.app/>
- **Parallel and Optimised R Programming:** single-day workshop, University of Southampton; delivered three times. <https://github.com/r-rse/parallel-r-materials>
- **Optimising R:** workshop for University of Sheffield collaborators working with the UK Office for Health Improvement and Disparities (OHID). <https://optimising-r.netlify.app/>
- **Introduction to R:** course for the Marine Biological Association.

RESEARCH SOFTWARE ENGINEER – UNIVERSITY OF SHEFFIELD – 31/05/2017 – 08/05/2022 – SHEFFIELD, UNITED KINGDOM

Member of the central Research Software Engineering team supporting researchers across the University to build robust analysis pipelines and software, promote best practice in research programming and digital resource management, and shift to a more open and collaborative research culture. Capacity building was a core part of the role, delivered through advocacy, code clinics and original training materials.

SELECTED SOFTWARE PROJECTS:

- **SeabirdDB:** R package and metadata webpage providing access to a database of seabird diets collected around the British Isles, developed from client-supplied data. <https://github.com/annakrystalli/seabirdDB> · <https://annakrystalli.me/seabirdDB/>
- **SedMaps:** Shiny app for visualisation, extraction and spatial selection of raster maps of the north-west European Shelf sedimentary environment, supporting both pre-built administrative polygons and custom-drawn extraction polygons. <https://github.com/annakrystalli/sedMaps>
- **emodnet.wcs:** R package providing access to EMODnet geographic vector data through EMODnet Web Feature Services. (Counterpart to the later emodnet.wcs package developed under R-RSE.) <https://github.com/EMODnet/EMODnetWFS>
- **CAMP SITE:** R package and Shiny app formalising client research code into installable, user-friendly software: simulation code refactoring and a paper-linked app for visualising replicate simulation results. <https://github.com/annakrystalli/CAMP SITE>
- **Research Software Engineering consultant for AMRC OWAST:** supported the Hole Generation group with their experimental, data-processing and reporting pipeline: designed a next-generation results database and ingress application; containerised and refactored a prototype Shiny app using the Golem framework.

SELECTED TRAINING MATERIALS:

- **Git & GitHub Through GitKraken Client — From Zero To Hero!:** version control course covering fundamentals and advanced team collaboration; delivered ten times. <https://srse-git-github-zero2hero.netlify.app/>
- **Introduction to GIS in R:** vector and raster spatial data in R. <https://annakrystalli.me/intro-r-gis/>
- **Introduction to Deep Learning in R:** R translation and delivery of an originally Python-based course co-developed with the RSE team; delivered three times. <https://rses-dl-course.github.io/>
- **Dataspice Tutorial:** introduction to the dataspice R package for capturing dataset metadata; developed for EMODnet Biology Data Products. <https://annakrystalli.me/emodnet-dataspice-tutorial/>

See Conferences & Seminars for selected talks, advocacy and community-building work.

RESEARCH DATA SCIENTIST / RESEARCH SOFTWARE ENGINEER – FREELANCE – 31/10/2014 – 31/05/2017 – SHEFFIELD, UNITED KINGDOM

Provided analytical and code-development support in R to an international team of Principal Investigators across the Universities of Sheffield, Bath (UK), UNAM (Mexico) and Seville (Spain), supporting two related macroecology projects:

- **Evolution of Sex Roles in Birds**: compiled a 1.3M-data-point dataset (202 variables) on avian reproductive, morphological, life-history and ecological traits. Developed tools to clean, validate and explore the data; parallelised range-extraction scripts to run on HPC; built a [Shiny app](#) for interactive trait exploration where Rmarkdown reports proved too unwieldy. Code, data and analyses made public.
- **Bird Trait Correlation Networks**: used the dataset above to build phylogenetically corrected trait correlation networks (rnetcarto, igraph).

EDUCATION AND TRAINING

31/08/2010 – 30/04/2015 Sheffield, United Kingdom

PHD IN MARINE MACROECOLOGY University of Sheffield

Investigated the macroecological temporal dynamics of regional population structure of copepod species at varying temporal and spatial resolutions using generalised linear and mixed modelling. Developed an environmentally informed method of copepod data interpolation combining satellite imagery and machine learning. Included a week-long Advanced Statistics in R course in the Department of Animal & Plant Sciences.

Field of study Natural sciences, mathematics and statistics | **Level in EQF** EQF level 8 |

Thesis Abundance-occupancy relationships in North Sea copepods: ventures into time and space

Link <https://etheses.whiterose.ac.uk/13831/>

31/07/2007 – 31/05/2009 Plymouth, United Kingdom

BSC (HONOURS) IN MARINE BIOLOGY & OCEANOGRAPHY University of Plymouth

Achieved a 1st Class Honours Degree. Awarded Babbage Scholarship of £3000 for performance in 2nd year.

Field of study Biological and related sciences not elsewhere classified | **Final grade** 1 | **Level in EQF** EQF level 6 |

National classification Bachelors Degree in Science with Honours |

Thesis Long-term trends in abundance and phenology of North Sea Continuous Plankton Recorder (CPR) fish larvae in relation to environmental and trophic drivers

04/11/2018 – 18/02/2019

SOFTWARE CARPENTRY INSTRUCTOR TRAINING The Carpentries

Website <https://carpentries.org/become-instructor/>

06/2013

SPATIAL STATISTICS IN R British Ecological Society Macroecology special interest group workshop

03/2012 – 07/2012

MACHINE LEARNING Stanford University - Coursera

Website <https://www.coursera.org/learn/machine-learning>

SKILLS

Languages & Data

R | Python | SQL | Web: HTML, CSS | LaTeX (Basic)

R Ecosystem

Data manipulation tidyverse (including ggplot2) | R package development (CRAN) | Quarto | Advanced RMarkdown | R Shiny | targets | renv | Positron IDE | Rstudio

Reproducible Research & Infrastructure

Git | GitHub | Github Actions | Docker | Scientific Writing & Reproducible Research: LaTeX, Jupyter, Rmd | CI/CD | Research Data Management | Containerised computational environments

Modelling & Analysis

Data Science | Statistical programming | Spatial Analysis | Modeling hubs | Machine Learning | Deep Learning | Keras | Forecast evaluations | Ensembling

AI-augmented development

Claude Code | Prompt engineering | AI skill authoring | Repository-level AI configuration | Structured knowledge bases for AI coding assistants

HONOURS AND AWARDS

10/04/2019

Software Sustainability Fellow 2019 – Software Sustainability Institute

Awarded £3,000 to pursue development of the ReproHack project, one day reproducibility hackathons aiming to help author and participating researchers learn about reproducibility through hands on reviewing and attempting to reproduce papers from associated published code and data.

Link <https://software.ac.uk/blog/2019-04-11-announcing-2019-software-sustainability-institute-fellows>

17/09/2019

Best Poster at 4th Conference of Research Software Engineering (2019) – Society of Research Software Engineering

Poster on pkgreviewr, an R package supporting rOpenSci package reviewers. Prize sponsored by Microsoft and included mentorship sessions at the Microsoft Cambridge offices.

Links <https://rseconuk2019.sched.com/event/SMZv/p17-pkgreviewr-supporting-ropensci-package-reviews-through-guidance-automation-and-templating> | <https://www.microsoft.com/en-us/research/blog/inclusive-environments-empower-big-dreams/>

25/06/2017

Network 50 – Mozilla

Named one of 50 People Who Made the Internet a Better Place in 2016.

Links <https://foundation.mozilla.org/en/blog/network-50/> | <https://storyengine.io/anna-krystalli/>

PROJECTS

23/11/2016 – CURRENT

ReproHack Project

Reproducibility hackathons where participants attempt to reproduce published research using publicly available code and data, providing structured feedback to authors. Started 2016 as part of OpenCon satellite events (Berlin, London 2017) and developed further through my 2019 Software Sustainability Institute Fellowship. Additional funding from [N8 CIR](#) and [EPSRC](#) supported extension events (including the [N8 Northern Tour Series](#) and HPC ReproHack at the University of Warwick). I am project lead and main developer of the [ReproHack Hub](#) (Django/Python), the online platform that lets anyone run their own ReproHack event.

Link <https://www.reprohack.org/>

CONFERENCES AND SEMINARS

06/10/2025 – 08/10/2025 Philadelphia, USA

USRSE 25: The Hubverse: Streamlining Collaborative Infectious Disease Modeling for Public Health Impact

Talk introducing the hubverse generalised forecast modelling hub infrastructure to the wider Research Software Engineering community: motivation, architecture, current deployments and contribution opportunities. Write here the description

Link <https://hubverse-org.github.io/hubverse-talk-usrse25/#/title-slide>

08/02/2022 Online, First AstraZeneca (AZ) R Conference 2022

Panel discussion: Reproducibility in R

Panelists: Anna Krystalli (University of Sheffield RSE), Diane Beldame (Thinkr), Abhijit Dasgupta (AZ, Machine Learning & AI) and Damian Zajac (AZ, Centralized Monitoring).

24/09/2021 Online, MetaScience 2021 Conference

Panel: Engaging the community in research reproduction

Panel discussion on community engagement in reproducing published research, drawing on my work organising ReproHack. Recording on the Centre for Open Science YouTube channel.

Links <https://metascience2021.org/events/engaging-the-community-in-research-reproduction/> | https://www.youtube.com/embed/nUWFYSF_I9k

15/12/2020 Online, University College London Tech Socials Series

Putting the R into Reproducible Research: Director's Cut

Whistle-stop tour of the breadth of available tools, demonstrating the ways R and the Rstudio integrated development environment can be used to underpin more open reproducible research and facilitate best practice.

Links <https://annakrystalli.me/talks/r-in-repro-research-dc-ucl.html#1> | <https://www.eventbrite.co.uk/e/kq-codes-technical-social-wed-16th-december-2020-anna-krystalli-tickets-131359733499>

09/07/2020 Online, useR 2020 Conference

useR 2020 Keynote: Computational Reproducibility: from theory to practice

Drew on experience as a Research Software Engineer, rOpenSci package editor and ReproHack co-founder to take stock of community progress on reproducibility and identify outstanding gaps. Slides and recording available.

Links <https://www.youtube.com/embed/KHmW8fV2NXo> | <https://annakrystalli.me/talks/user2020.html#1> | <https://user2020.r-project.org/program/keynotes/>

08/07/2019 British Ecological Society Quantitative & Movement Ecology SIG join meeting

What can RSEs do for U?

Invited talk at the BES Quantitative & Movement Ecology 2019 Annual Meeting to speak about what Research Software Engineers do and how they can support academics with their computational research.

Link https://annakrystalli.me/talks/bes_sig19.html

17/03/2019 University of Sheffield

The Turing Way: Build a BinderHub

Was part of the Turing Way project team who organised and facilitated a Build a BinderHub Workshop during which we demonstrated how to build a BinderHub on Microsoft Azure cloud computing resources.

Link <https://rse.shef.ac.uk/training/workshop/2019-03-18-build-a-binderhub/>

20/05/2018 – 21/05/2018 Microsoft Reactor, Seattle, Washington, USA

rOpenSci Unconf 2018

rOpenSci Unconf 2018. Proposed and led work on minimal dataset metadata standards, files and formats, which became the dataspace R package (now on CRAN; see Creative Works).

Links <https://unconf18.ropensci.org/> | <https://github.com/ropensci/unconf18/issues/72> | <https://github.com/ropensci/dataspace>

03/08/2017 – 05/08/2017 GooglePlex, Mountain View California, USA

Science Foo (SciFoo) 2017

Invited unconference (O'Reilly Media / Digital Science / Alphabet Inc.). Proposed and chaired session: *Reproducible Research: Everyone wants it but who is going to teach us to do it?!*

Links https://en.wikipedia.org/wiki/Science_Foo_Camp | <https://www.digital-science.com/blog/2017/08/collective-noun-science-scifoo-2017/>

● NETWORKS AND MEMBERSHIPS

21/06/2018 – 07/09/2024

rOpenSci Software Review Editor

Editor for rOpenSci's software peer review system from May 2018 to September 2024. rOpenSci's community-driven peer review of R packages contributed to the scientific R ecosystem. Editorial work included triaging submissions, recruiting reviewers, and shepherding packages through review to acceptance.

Link <https://ropensci.org/>

31/03/2016 – 26/02/2022 Sheffield, UK

Sheffield R Users Group

The Sheffield R Users hosted monthly talks on all topics related to R. I gave a number of talks and demos and was also co-organiser of the group from 2016 to 2020.

Link <https://www.meetup.com/SheffieldR-Sheffield-R-Users-Group>

07/02/2021 – 26/05/2021

Open Life Sciences

The Open Life Science (OLS) program helps **individuals and stakeholders in research** to become **Open Science ambassadors** and is in many ways an evolution of the sun-setted Mozilla Science Lab initiative. I participated as a mentor in OLS Cohort 3.

Link <https://openlifesci.org/ols-3>

01/01/2018 – 30/12/2020

The Turing Way

Member of the small launch team that established **The Turing Way**, The Alan Turing Institute's open-source, community-driven handbook for reproducible, ethical and collaborative data science. Hired through the University of Sheffield Research Software Engineering team to support the project's launch and early development; contributions spanned book content, community workshops (e.g. *Build a BinderHub* at the University of Sheffield, March 2019 — see Conferences & Seminars), and early community-building. The handbook has since grown into one of the largest community-driven open-source projects in reproducible and ethical data science, with hundreds of contributors worldwide.

Link <https://the-turing-way.netlify.app/welcome>

31/12/2015 – 31/08/2018

Mozilla Science Lab

Member of the inaugural cohort of Mozilla Science Lab's Open Leadership programme (Working Open Workshop, Berlin 2016). Continued involvement as a mentor for subsequent cohorts, ran a Global Sprint site at the University of Sheffield (2017, 2018), and facilitated sessions at MozFest London (2016, 2017, 2019) including *Git & GitHub through Rstudio* in 2016.

Link <https://wiki.mozilla.org/ScienceLab>

CREATIVE WORKS

01/09/2023 – CURRENT

hubverse R package ecosystem

Major contributor and maintainer across the **hubverse** open-source R package ecosystem: a suite of community-maintained R packages providing collaborative modelling-hub infrastructure used by ~two dozen modelling hubs internationally, including hubs hosted and/or used by the US Centers for Disease Control and Prevention, the European Centre for Disease Prevention and Control, the Australia-Aotearoa Consortium for Epidemic Forecasting and Analytics, and the California Department of Public Health. Lead or substantial contributor across packages; hubUtils, hubValidations, hubData, hubAdmin, hubEvals, hubCI and the hubverse dashboard infrastructure.

Links <https://github.com/hubverse-org> | <https://hubverse.io>

01/08/2022 – 31/10/2022

emodnet.wcs R package

R package providing access to the EMODnet Biology Web Coverage Service for retrieval, filtering and analysis of gridded environmental and biological data in R. Counterpart to the earlier emodnet.wcs package. Developed as part of EMODnet Biology project phases.

Link <https://emodnet.github.io/emodnet.wcs/>

30/04/2020

emodnet.wfs R package

Package developed as part of EMODnet Biology Phases III & IV to enable interrogation, filtering and access of EMODnet Web Feature Service data through R.

Links <https://github.com/EMODnet/EMODnetWFS/> | <https://emodnet.github.io/EMODnetWFS/>

31/03/2018

dataspice R package

R package making it easier to create basic, lightweight, concise metadata files for tabular datasets by editing CSVs. Built around Schema.org/Dataset and related metadata standards. Project initiated at rOpenSci Unconf 2018; now on CRAN.

Links <https://github.com/ropensci/dataspice> | <https://docs.ropensci.org/dataspice/> | <https://annakrystalli.me/emodnet-dataspice-tutorial/>